

Leakage-Controlled and Deployment-Oriented Diagnosis of Input-Resolution Trade-offs for Dense Broiler Detection

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Keypoints/Highlights

- A source-aware, leakage-controlled protocol was developed for dense broiler imagery.
- Input-resolution trade-offs were diagnosed using detection, counting, uncertainty, and deployment cost.
- No single resolution was uniformly preferable across detection, counting, and cost objectives.
- Leakage-cleaned PigDetect validation confirmed strict-localization gains with increasing resolution.

Impact

This study provides an auditable basis for diagnosing detector input-resolution trade-offs in dense poultry monitoring by linking strict-localization accuracy, threshold-calibrated counting, and hardware demand. The framework supports selection of objective-specific operating points for computer-vision systems in precision livestock farming.

Abstract

Purpose Input resolution strongly affects dense broiler detection, but conclusions can be misleading when data leakage, counting calibration, and deployment cost are considered separately. This study developed a leakage-controlled, deployment-oriented protocol for diagnosing input-resolution trade-offs in precision poultry monitoring.

Methods A processed PIO dataset (1,487 images; 327,288 instances) was split using exact-duplicate removal, source-group binding, density-aware allocation, and cross-split audits. YOLO11n was evaluated at 800, 960, and 1280 pixels under a fixed protocol. Comparisons combined detection accuracy, scale- and density-stratified analyses, validation-calibrated counting, paired bootstrap uncertainty, and computational cost. Independent external validation used a leakage-cleaned PigDetect split.

Results On PIO, test mAP50-95 increased from 0.7199 at 800 pixels to 0.7420 at 960 and 0.7543 at 1280. Under the adopted validation-based threshold-transfer protocol, the 800-pixel model achieved the lowest test counting MAE. Under the specified illustrative composite weighting, 960 pixels obtained the highest composite score, whereas 1280 pixels achieved the highest box-level accuracy. On PigDetect, mAP50-95 increased from 0.7673 to 0.7757 and 0.7844, with paired 95% confidence intervals for both adjacent gains above zero. Higher resolution improved strict localization but increased computation and memory demand.

Conclusion No single resolution was uniformly preferable across all objectives. In the tested setting, 800 pixels minimized computational demand and, under the adopted threshold-transfer protocol, produced the lowest test counting MAE; 960 pixels ranked highest only under the specified illustrative composite weighting; and 1280 pixels achieved the highest box-level accuracy. PigDetect showed the same resolution-dependent strict-localization trend.

Keywords: Precision poultry monitoring; Broiler chicken detection; Input resolution; Leakage-controlled evaluation; Deployment-oriented diagnosis; Counting calibration

Introduction

Vision-based sensing has become an important tool for precision livestock and poultry monitoring because it can provide non-contact information about animal distribution, flock density, and production-environment status in commercial houses (Guo et al. 2023; Guo et al. 2025; Li 2025; Boniche et al. 2026; Bhujel et al. 2025). In broiler production, reliable image-based detection is a prerequisite for automated counting, density assessment, welfare-related observation, and real-time monitoring. However, commercial broiler scenes are visually challenging: birds are densely distributed, visually similar, frequently occluded, and often represented by small bounding boxes under overhead or oblique camera views. These properties make detection and counting sensitive to input sampling resolution, feature-map representation, non-maximum suppression, and computational constraints. Therefore, input resolution is not only a detector hyperparameter, but also a practical design variable for precision poultry monitoring systems that must balance detection accuracy, counting reliability, and deployment cost.

Input resolution is one of the most direct but often under-diagnosed factors affecting dense small-object detection. Increasing input resolution can preserve more spatial detail and may improve localization for small or crowded objects. However, higher resolution also increases computational cost, GPU memory consumption, and training time. More importantly, the benefit of higher resolution is not necessarily uniform across object scales, scene densities, or monitoring objectives. A setting that improves global mean average precision may not produce the lowest counting error after confidence-threshold transfer, while a lower resolution may already capture most of the practically useful gain under deployment constraints. Input-resolution analysis should therefore be framed as a diagnosis of objective-specific trade-offs rather than a search for a single best setting.

Another critical issue is evaluation leakage. Dense image datasets collected from similar viewpoints, video streams, or repeated acquisition conditions may contain duplicate or near-duplicate images across training, validation, and test splits. If such leakage is not controlled, the resulting performance estimates can be overly optimistic and may not reflect true generalization. Similarity-aware splitting and leakage-reduced evaluation have been emphasized in recent machine-learning studies (Joeres et al. 2025), and image-similarity tools such as SSIM and perceptual hashing are useful for auditing near-duplicate visual samples (Wang et al. 2004; Biswas and Blanco-Medina 2021). Paired bootstrap analysis can further support uncertainty estimation when comparing resolution-dependent AP differences (Efron and Tibshirani 1993). A reliable evaluation protocol should therefore combine duplicate removal, source-aware grouping, split-level audit checks, and transparent release of split metadata.

This study develops a leakage-controlled and deployment-oriented framework for diagnosing input-resolution trade-offs in dense broiler monitoring. Input resolution is treated as an application-level design variable that links image sampling, small-object representation, counting calibration, and hardware demand. The framework is intended to turn resolution configuration from an informal training choice into an auditable selection of an objective-specific operating point supported by scene-specific accuracy, uncertainty, and deployment evidence.

The framework combines MD5-based duplicate removal, source-group binding, density-balanced split allocation, source-component diagnostics, and multiple cross-split leakage checks before resolution effects are measured. The prespecified primary grid of 800, 960, and 1280 pixels spans low-, intermediate-, and high-compute operating regimes under the fixed 8 GB GPU and batch-size constraint. A 1024-pixel run was

subsequently added as an exploratory intermediate-resolution sensitivity check between 960 and 1280 pixels without changing the complete primary experiment matrix.

The study makes three application-oriented contributions. First, it adapts evaluation to dense poultry imagery through auditable source-aware splitting, near-duplicate screening, density balancing, and maximum-detection verification. Second, it integrates global AP, scale- and density-stratified AP, SGF-AP, TP/FN/FP diagnostics, validation-calibrated counting, paired uncertainty analysis, and deployment cost into a unified input-resolution diagnosis workflow. Third, it tests the resulting resolution pattern through multi-seed PIO experiments, an auxiliary YOLO-family cross-version robustness check, and independent livestock-domain validation on leakage-cleaned PigDetect. Together, these elements provide a practical methodology for interpreting objective-specific operating points around accuracy, counting, and deployment requirements.

The resulting workflow proceeds from auditable data preparation to controlled resolution comparison, subgroup diagnosis, counting calibration, uncertainty estimation, and deployment-oriented interpretation. This structure allows each reported operating point to be traced to a stated monitoring objective and to the measured behavior of dense livestock scenes.

The results section reports global, scale-stratified, density-stratified, counting, uncertainty, deployment, YOLO-family cross-version, and independent PigDetect livestock-domain validation analyses.

Poultry and livestock vision in dense monitoring scenes

Computer vision has become an important sensing modality in precision livestock and poultry monitoring because it can support non-contact observation of animal distribution, activity, resource use, and flock-level production status (Guo et al. 2023; Guo et al. 2025; Li 2025; Boniche et al. 2026; Bhujel et al. 2025). In commercial broiler houses, image-based detection is a prerequisite for automated counting, density assessment, welfare-related observation, and real-time decision support. However, poultry scenes are more challenging than many conventional agricultural vision tasks because animals are visually similar, densely distributed, partially occluded, and often captured under variable illumination and camera viewpoints. These conditions make detection performance sensitive to object scale, scene density, annotation quality, and deployment constraints.

Previous broiler and livestock studies have used YOLO-style detection, attention mechanisms, density-map counting, and tracking tools to address animal detection, behavior monitoring, flock counting, and welfare-related analysis (Guo et al. 2023; Guo et al. 2025; Neethirajan 2022; van der Eijk et al. 2022; Lv et al. 2023; Guo et al. 2022). Public datasets and benchmark resources for precision poultry and livestock farming have also improved the reproducibility of vision-based monitoring studies (Li 2025; Boniche et al. 2026; Bhujel et al. 2025). The PIO dataset provides a large-scale open benchmark for broiler chicken detection under real poultry farming conditions (Boniche et al. 2026). Compared with prior studies that mainly focus on model design, behavior recognition, or counting architecture, the present study focuses on a different but practically important question: how input-resolution trade-offs should be diagnosed and translated into objective-specific operating points when data leakage, scene density, counting accuracy, uncertainty, and deployment cost are jointly considered.

Dense small-object detection and input-resolution effects

Dense broiler detection can also be viewed as a dense small-object detection problem. General object detection has progressed from region-proposal methods to efficient one-stage and multi-scale detectors (Ren et al. 2017; Liu et al. 2016; Lin et al. 2014; Lin et al. 2017; Tan et al. 2020; Bodla et al. 2017; Padilla et al. 2021). However, dense small-object detection remains difficult because small targets provide limited spatial evidence, are easily confused with background clutter, and are strongly affected by occlusion, scale variation, and non-maximum suppression behavior (Nikouei et al. 2025; Feng et al. 2023; Cheng et al. 2023). In high-density poultry images,

these challenges are amplified because hundreds of visually similar animals may appear in a single image, and many individual birds occupy only a limited number of pixels.

Multi-scale feature pyramids, compound scaling, slicing-aided inference, and high-resolution imagery benchmarks have been widely used to improve representation and localization for small objects (Lin et al. 2017; Tan et al. 2020; Akyon et al. 2022; Li et al. 2020). These studies motivate the need to analyze not only global AP, but also scale-stratified and density-stratified behavior when changing input resolution. The present study differs from slicing-aided or multi-scale inference methods that modify how images are processed at inference time. Instead, the detector architecture, training configuration, data split, and test rules are kept fixed, and input resolution is treated as a controlled design variable. This allows resolution-dependent gains to be interpreted in terms of detection accuracy, counting behavior, uncertainty, and deployment cost rather than being reported as a simple hyperparameter search result.

Leakage-controlled evaluation, similarity audit, and uncertainty

Reliable evaluation is essential for precision agriculture vision systems because over-optimistic performance estimates can lead to incorrect deployment decisions. In image datasets collected from similar camera viewpoints, video streams, repeated sampling conditions, or source-related acquisition sessions, duplicate, near-duplicate, or source-related samples may appear across training, validation, and test subsets. If such leakage is not controlled, the test set may no longer represent truly unseen data, and reported detection or counting performance may be inflated (Joeres et al. 2025).

Image-based leakage control can be supported by exact-duplicate removal, source-group binding, perceptual-hash candidate screening, and SSIM-based near-duplicate verification (Wang et al. 2004; Biswas and Blanco-Medina 2021). For dense poultry scenes, source-level grouping is especially important because images collected from the same house, camera, or acquisition sequence may share strong visual similarity even when they are not exact duplicates. In addition, paired bootstrap analysis provides a practical way to quantify uncertainty in resolution-dependent AP differences while preserving image-level coupling among dense predictions (Efron and Tibshirani 1993). Therefore, an input-resolution diagnosis study should combine leakage-controlled split construction, cross-split similarity audit, and uncertainty analysis before drawing conclusions about the benefit of a particular input resolution.

YOLO-style detection and deployment constraints for precision poultry monitoring

YOLO-style detectors are widely used in real-time object detection because they provide a practical balance between accuracy and speed (Redmon et al. 2016; Redmon and Farhadi 2017, 2018; Bochkovskiy et al. 2020; Wang C.-Y. et al. 2021, 2023; Wang C.-Y. et al. 2024; Wang A. et al. 2024; Jocher and Qiu 2024). This balance is particularly relevant for precision poultry monitoring, where models may need to operate under limited GPU memory, edge-computing resources, or near-real-time farm-monitoring requirements. Later YOLO variants improved feature aggregation, training strategies, and detector efficiency, while recent software implementations such as Ultralytics YOLO11 further support practical deployment workflows (Jocher and Qiu 2024).

For dense broiler scenes, increasing input resolution can preserve more spatial detail and improve localization for small or crowded birds. However, higher resolution also increases GFLOPs, latency, memory pressure, and training time. Input resolution should therefore not be treated as a purely accuracy-oriented decision. A setting that maximizes mAP may not be preferable for scalar counting, low-cost deployment, or real-time monitoring. This motivates the leakage-controlled and deployment-oriented input-resolution diagnosis protocol developed in this study.

Materials and methods

Overview of the leakage-controlled input-resolution diagnosis protocol

This study implemented a scene-adapted resolution diagnosis framework for dense broiler detection in precision poultry monitoring. The workflow controls the detector architecture and optimization protocol while explicitly addressing duplicate leakage, source imbalance, object-scale variation, scene density, confidence-threshold calibration, statistical uncertainty, and computational cost. These adaptations align the evaluation with the practical conditions of high-density poultry imagery and farm-side deployment.

Given an image I_i and a primary input resolution r in $\{800, 960, 1280\}$, the detector produces a set of bounding-box predictions $P_{i,r}$ with confidence scores and class labels. The same detector architecture, training protocol, data split, and evaluation rules were used across all primary resolutions, making input resolution the controlled design variable. The 1024-pixel setting was introduced after the primary matrix as an exploratory intermediate-resolution sensitivity check; it was used to inspect the intermediate region but was not added retrospectively to the full seed-repeatability, subgroup, bootstrap, and deployment experiment matrix.

Fig. 1 summarizes the controlled front end of the input-resolution diagnosis framework. The protocol defines the dense PIO monitoring case and its object-density statistics, constructs an audited grouped train/validation/test split, and evaluates the 800-, 960-, and 1280-pixel primary operating points under the same YOLO11n architecture, split, training protocol, and test rules. The subsequent analyses connect these controlled runs to global accuracy, scale and density behavior, SGF-AP, bootstrap uncertainty, validation-calibrated counting, deployment cost, and an illustrative task-objective score. The 1024-pixel result is retained only as an exploratory sensitivity probe around the intermediate operating region.

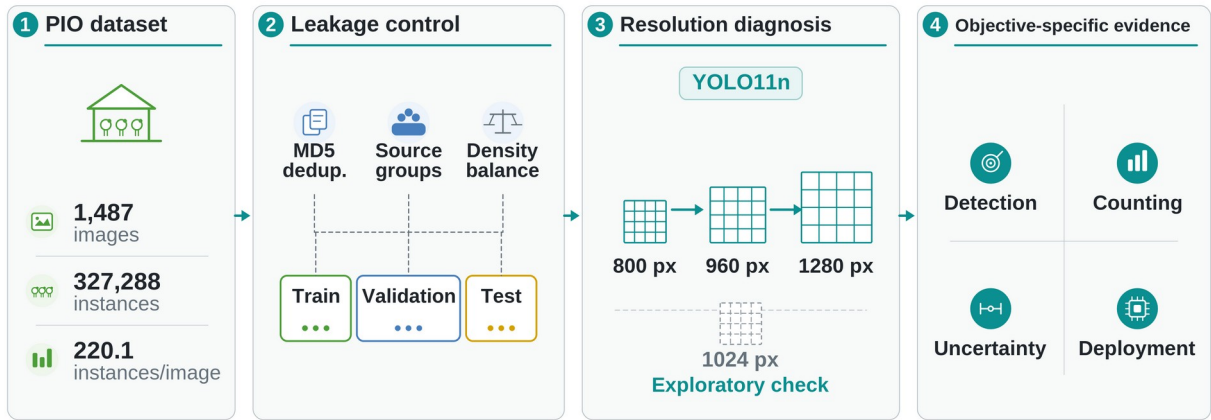


Fig. 1 Leakage-controlled experimental framework for diagnosing input-resolution trade-offs in dense broiler monitoring. Stages 1-4 summarize the PIO dataset and density statistics, leakage-control and grouped splitting procedure, controlled comparison of the prespecified 800-, 960-, and 1280-pixel primary operating points under a fixed YOLO11n protocol, and the objective-specific evidence groups used for detection, counting, uncertainty, and deployment diagnosis. The 1024-pixel setting is shown only as an exploratory intermediate-resolution sensitivity check and was not included in the complete primary experiment matrix.

Dense broiler-monitoring case study and leakage-controlled split

The PIO dataset (Boniche et al. 2026) was used as a real-world dense broiler-monitoring case study for precision poultry farming. The published data descriptor reports 1,487 images and 327,289 broiler instances. Direct counting of the label files used in this study yielded 327,288 annotation records. The origin of this one-record discrepancy could not be reconstructed from the available source and preprocessing records; therefore, all dataset statistics and experiments reported here use the frozen label files.

The fixed split, denoted as PIO-GRDB-MD5-7_1_2, contains 1,027 training images, 153 validation images, and 307 test images. The corresponding annotation-record counts are 225,771, 36,905, and 64,612 for training, validation, and testing, respectively, with 327,288 records in total and an average of 220.10 records per image. Direct comparison with the available source copy and inspection of the Ultralytics data caches confirmed that split construction introduced no change in image or annotation-record counts. Five source records had zero normalized width and were retained unchanged in the source copy, frozen split, and data cache; no post hoc label correction was performed after the experimental protocol was frozen.

The original PIO folder names contain prefixes such as train__ and val__ inherited from the source collection; these prefixes do not reflect the leakage-controlled split assignment used in this study.

The split protocol was designed to reduce the risk of over-optimistic evaluation caused by duplicate images, near-duplicate images, or source-level overlap. Image-level MD5 hashing was first used to remove exact duplicates. Source groups were inferred from source-path and filename-pattern metadata and were treated as indivisible units during split construction so that near-contiguous or source-related samples were not distributed across training, validation, and test subsets. A density-balanced greedy allocation strategy was used to reduce severe distributional imbalance in the number of instances per image across subsets.

Four cross-split leakage checks were performed using original image paths, original file names, image-content MD5 values, and inferred source-group identifiers. In addition, near-duplicate screening was conducted using perceptual-hash candidates followed by structural similarity checks on 256×256 grayscale images (Wang et al. 2004). No confirmed cross-split duplicate or near-duplicate image pairs were found under the adopted audit criteria. The full leakage-control and near-duplicate audit configuration is provided in Online Resource 1 (Table S3a), and the grouped split procedure is described in Algorithm 1.

Source-component diagnostics are provided in Online Resource 1 (Table S3). At the source-component level, the final split contained 361 training components, 2 validation components, and 20 test components. The largest components contained 352, 130, and 151 images in the training, validation, and test subsets, respectively. These diagnostics expose source-group concentration and split imbalance transparently rather than treating the split as a random image-level partition. The validation subset was used for model monitoring and confidence-threshold calibration, whereas the test subset was reserved for test-set performance reporting and diagnostic analysis. The dataset is therefore used as a controlled dense broiler-monitoring benchmark rather than as the primary contribution of the study.

Table 1 summarizes the image and annotation-record distributions of the leakage-controlled training, validation, and test subsets.

Table 1 Dataset split statistics under the PIO-GRDB-MD5-7_1_2 protocol

Split	Images	Image ratio (%)	Instances	Instance ratio (%)	Mean inst./image
Train	1,027	69.07	225,771	68.98	219.84
Validation	153	10.29	36,905	11.28	241.21
Test	307	20.65	64,612	19.74	210.46
Total	1,487	100.00	327,288	100.00	220.10

Algorithm 1 Leakage-controlled grouped split construction

Input: processed image set I , annotation set A , target split ratio 7:1:2, source-path and filename metadata M , image-level density values, and leakage keys {source path, filename, image-content MD5, source group}.
Output: leakage-controlled train, validation, and test split manifests.

1. Compute the image-content MD5 hash for each image and remove exact duplicates.

2. Infer a source-group identifier for each remaining image using source-path and filename-pattern metadata.
3. Construct grouped units so that all images in the same source group remain together.
4. For each group, compute the number of images, number of annotated instances, mean density, and density-bin composition.
5. Sort grouped units by size and density characteristics.
6. Assign grouped units to train, validation, or test using greedy allocation while minimizing deviations from the target image ratio, target instance ratio, and density-bin balance.
7. Compute split-level image, instance, and density statistics.
8. Perform cross-split leakage checks using source path, filename, image-content MD5, and source group.
9. Perform near-duplicate audit using $pHash \leq 6$ or $dHash \leq 6$ candidate screening followed by $SSIM \geq 0.95$ verification on 256×256 grayscale images.
10. If any cross-split leakage or confirmed near-duplicate pair is found, revise the grouped allocation and repeat the audit.
11. Freeze the split manifest as PIO-GRDB-MD5-7_1_2.

Detector configuration for dense broiler detection

YOLO11n was selected as the main detector because its lightweight architecture is well suited to deployment-oriented dense livestock monitoring, where inference speed, memory footprint, and training cost are practical constraints (Jocher and Qiu 2024). Keeping the detector configuration fixed allowed the study to concentrate on scene-specific resolution effects and on the evaluation adaptations required for dense poultry images. YOLOv8n was additionally evaluated as an auxiliary YOLO-family cross-version robustness check.

Three primary input resolutions were prespecified: 800, 960, and 1280 pixels. They were chosen to represent clearly separated low-, intermediate-, and high-compute operating regimes while remaining compatible with the detector stride and a common batch size of 2 on the available 8 GB GPU. Relative to 800 pixels, 960 pixels increases the image side length by 20% and the pixel area by 44%, providing a distinct intermediate workload without approaching the substantially higher 1280-pixel cost. The 1024-pixel run was added after the main matrix as an exploratory intermediate-resolution sensitivity check between 960 and 1280 pixels. Because it did not receive the complete multi-seed, subgroup, paired-bootstrap, and repeated-deployment evaluation assigned to the three prespecified candidates, it was retained as supporting evidence rather than promoted to a fourth primary operating point.

All primary PIO YOLO11n models were trained for 300 epochs with batch size 2, stochastic gradient descent optimization, cosine learning-rate scheduling, and pretrained weights. Batch size 2 was retained across all resolutions to keep the optimization protocol fixed under the available 8 GB GPU constraint and to avoid confounding the resolution comparison with resolution-dependent batch-size changes. Lower resolutions may permit larger batch sizes on the same hardware, but evaluating batch-size-dependent optimization was outside the scope of the controlled input-resolution diagnosis protocol. The main YOLO11n experiments used seed 42. To examine whether the observed resolution trend was caused by a single random initialization or training trajectory, additional YOLO11n seed-repeatability experiments were conducted for all three resolutions using seeds 123 and 2024. The same training split, validation split, augmentation pipeline, optimizer configuration, and evaluation protocol were maintained across resolutions.

Detection evaluation and maximum-detection sensitivity analysis

Standard detection evaluation was first performed on the fixed test split. The primary metrics were precision, recall, mAP50, and mAP50-95. Since each image in the dataset may contain hundreds of object instances, the maximum number of retained detections per image is a critical evaluation

parameter. A low maximum-detection setting can artificially suppress recall in high-density scenes and lead to an underestimation of detection performance.

Therefore, the main test evaluation used $\text{max_det} = 1000$. Additional sensitivity analysis was conducted using lower and higher maximum-detection settings to verify whether the main results were limited by the detection cap. This analysis was used to confirm that the selected maximum-detection setting was sufficient for the density range of the test split.

Scale- and density-stratified evaluation of resolution effects

To identify where resolution-related performance gains originate, the test split was analyzed using both object-scale and image-density groupings.

For scale-stratified diagnosis, ground-truth boxes in the test split were divided into four groups according to the quartiles of the square root of bounding-box area. These groups are denoted as tiny, small, medium, and large. This quartile-based design ensures that the number of instances in each scale group is approximately balanced, preventing a dominant scale group from overwhelming the diagnostic analysis.

For density-stratified diagnosis, test images were divided into four groups according to the number of annotated instances in the full image. These groups are denoted as low, medium, high, and ultra-high density. The density group is defined at the full-image level, even when qualitative visualization only displays a local crop for readability.

To avoid ambiguity between subgroup-local evaluation and global contribution analysis, this study distinguishes ordinary subgroup AP from strict global-first diagnostic AP, hereafter denoted as SGF-AP.

Ordinary subgroup AP is computed after filtering the evaluation set to a predefined subgroup, such as a target scale group or a density group. Predictions and ground-truth objects outside the subgroup are excluded before forming the precision-recall curve. Therefore, ordinary subgroup AP answers the question of how well the detector performs when that subgroup is evaluated as a separate subset.

In contrast, SGF-AP preserves the full-test prediction order before subgroup attribution. Predictions are ranked and matched globally, and matched true positives are then assigned to their predefined scale or density groups. The ranking is not re-optimized within each subgroup.

SGF-AP therefore answers a complementary question: under the detector's global operating order, how much does a subgroup contribute to the overall ranked detection result? It is contribution-oriented, not numerically interchangeable with ordinary subgroup AP, and is used only as an auxiliary diagnostic metric. Additional implementation details and supporting tables are provided in Online Resource 1 (Tables S5-S7 and Fig. S2).

Bootstrap uncertainty analysis

Paired image-level bootstrap analysis was used to assess the stability of resolution-dependent performance differences (Efron and Tibshirani 1993). The image was used as the resampling unit because predictions and ground-truth annotations within the same image are statistically coupled, especially in dense scenes. For each bootstrap replicate, the same resampled image set was applied to all resolution settings, allowing paired comparisons between resolutions.

The bootstrap analysis estimated confidence intervals for AP differences between 800 and 960, between 960 and 1280, and between 800 and 1280. This paired design reduces the influence of image-

composition variability and provides a more stable estimate of whether observed resolution gains are consistent across the test set.

Counting calibration for broiler detection outputs

In dense small-object applications, detection accuracy and counting accuracy are related but not equivalent. Average precision is a ranking-based detection metric, whereas image-level counting error depends on a fixed confidence threshold and on the calibration of predicted scores. Therefore, counting performance was evaluated separately from AP.

A validation-calibrated counting protocol was used. For each resolution, a range of confidence thresholds was evaluated on the validation split, and the threshold with the lowest mean absolute error was selected, with RMSE and absolute bias used as auxiliary criteria when necessary. The selected threshold was then fixed and applied to the test split. The reported counting metrics included MAE, RMSE, MAPE, bias, and relative error.

The validation subset contained two source components, with the largest component contributing 130 of 153 validation images. This high source-component concentration means that the calibrated confidence threshold may be tuned partly to the density distribution of the dominant validation component rather than to the full range of possible validation sources. Threshold calibration followed the standard validation protocol, and sensitivity to threshold choice was additionally examined through the test-set diagnostic confidence sweep and the validation-source-component sensitivity check in Online Resource 1 (Tables S10 and S18-S19; Fig. S3).

A test-set confidence sweep was used only as a diagnostic sensitivity analysis and was not used to select the reported counting threshold. This separation prevents threshold selection from being tuned directly on the test set and provides a more realistic estimate of counting behavior under deployment-like conditions.

Deployment-oriented benchmarking for precision poultry monitoring

To evaluate the practical cost associated with the objective-specific operating points, deployment-oriented benchmarking was conducted under the same hardware and software environment for all resolutions. All training and benchmarking experiments were performed on a single NVIDIA GeForce RTX 5060 Laptop GPU with 8 GB of dedicated GPU memory (NVIDIA Corporation, Santa Clara, CA, USA). The repeated locked-weight deployment benchmark used the validation split, batch size 2, workers = 0, confidence threshold = 0.001, NMS IoU = 0.70, max_det = 1000, GPU FP16 inference, five warm-up repeats, and thirty formal repeats. Median preprocessing time, inference time, postprocessing time, total latency, latency IQR, GFLOPs, and validation/inference peak CUDA memory were reported.

Peak memory in the deployment benchmark refers to validation/inference CUDA memory rather than training peak VRAM. Training time was reported separately as a development cost, whereas inference latency, GFLOPs, and memory pressure represent recurring deployment costs. The illustrative composite score combines these complementary factors only to demonstrate one explicit accuracy-cost decision scenario; it was not intended to define a universally optimal resolution. The separate component values allow readers to apply different engineering priorities.

Hardware-specific acceleration platforms such as TensorRT, ONNX Runtime, embedded GPUs, and optimized NMS implementations may change absolute latency values. Therefore, the deployment analysis is interpreted together with training time, GFLOPs, total latency, and memory pressure rather than using a single speed metric as the only deployment criterion.

Independent livestock-domain external validation on PigDetect

PigDetect was added as an independent livestock-domain external validation dataset to test whether the resolution pattern observed on the primary broiler dataset extended to a different species, acquisition setting, and annotation source (Henrich et al. 2026). The original annotations were converted to YOLO format, and the supplied training, validation, and test partitions were audited for cross-split leakage before model training. The initial audit identified 18 SSIM-confirmed near-duplicate pairs between the training and validation subsets involving nine validation images. Removing those images produced a clean split of 2,431 training images, 241 validation images, and 250 untouched test images; the test subset contained 5,436 annotated pig instances. Re-audit found zero cross-split filename, filename-stem, image-MD5, label-MD5, and SSIM-confirmed near-duplicate overlap.

YOLO11n was trained independently at 800, 960, and 1280 pixels using pretrained initialization, SGD, batch size 2, seed 42, a maximum of 300 epochs, and patience = 50. The clean split, architecture, optimizer, seed, augmentation settings, and validation-based checkpoint-selection rule were fixed across resolutions. The best.pt checkpoint from each run was evaluated once on the untouched test set. PigDetect served as an external livestock validation, while training-seed repeatability was evaluated through the dedicated three-seed PIO experiment.

Test evaluation used confidence = 0.001, NMS IoU = 0.70, max_det = 1000, batch size 2, workers = 0, no test-time augmentation, and FP32 evaluation. Uncertainty in resolution-dependent differences was estimated with 1,000 paired image-level bootstrap replicates. In every replicate, the same sampled test-image indices were used for all three locked checkpoints, and 95% percentile confidence intervals were calculated. This procedure estimates uncertainty associated with resampling the held-out test images conditional on the trained checkpoints; it does not estimate training-seed uncertainty.

PigDetect scale analysis defined target size as the square root of bounding-box area in original-image pixels and divided the 5,436 test instances into four equal-count groups using the 25th, 50th, and 75th percentiles. Density analysis defined image density as the number of annotated pigs per test image and used tie-preserving near-quartile groups, so group sizes could differ when image counts were tied. The same verified prediction caches and locked test protocol were used for all subgroup analyses. Deployment benchmarking used the validation split, batch size 2, FP16 inference, five discarded warm-up repeats, and thirty formal repeats; median preprocessing, inference, postprocessing, total latency, latency IQR, GFLOPs, and peak allocated CUDA memory were recorded under the same NVIDIA GeForce RTX 5060 Laptop GPU environment.

Qualitative evidence used the locked PigDetect qualitative image set and one fixed visualization protocol. The same image and crop were compared across 800-, 960-, and 1280-pixel checkpoints. Displayed predictions used confidence ≥ 0.50 , while local FN/FP diagnostics used confidence ≥ 0.25 and IoU ≥ 0.50 . These display settings were used only for readability and did not alter the formal test metrics.

Reproducibility materials

To support reproducibility and transparent evaluation, the manuscript-linked public release v1.2.1 is provided for this study (Song 2026). The release contains the PIO leakage-controlled split manifest, source-group metadata, leakage-audit summaries, group definitions, bootstrap outputs, counting-calibration outputs, supplementary source data, evaluation scripts, traceability logs, and fixed figure source data. It also contains the PigDetect clean-split audit, locked test metrics, scale and density source tables, paired-bootstrap summary, repeated deployment benchmark, source-to-manuscript traceability map, and the locked qualitative artwork and protocol metadata.

The PIO plotting scripts read values from fixed CSV files rather than hard-coded plotting arrays. Tables S1–S20, including Tables S3a, S4a–S4b, and S17a–S17g, in Online Resource 1 are linked to source records in `supplementary_source_data`; Table S21 is the reproducibility checklist. The PigDetect protocol and external-validation records are organized under `external_validation/PigDetect`. Raw PIO and PigDetect images, trained model weights, and prediction caches are not redistributed. These materials support inspection of the leakage-controlled splits, resolution-dependent analyses, subgroup evaluations, bootstrap procedures, deployment benchmarks, and fixed qualitative-evidence protocol wherever authorized access to the original datasets is available.

Results

Global detection performance across input resolutions

The standard test-set results and paired mAP50-95 uncertainty are summarized in Table 2. The evaluation used the fixed test split, confidence threshold of 0.001, NMS IoU of 0.70, and `max_det` = 1000.

Table 2 Global test-set detection performance and paired mAP50-95 uncertainty

Resolution	Precision	Recall	mAP50	mAP50-95	Bootstrap mean Δ vs. previous [95% CI]
800	0.9301	0.9283	0.9643	0.7199	—
960	0.9314	0.9385	0.9739	0.7420	+0.0195 [0.0167, 0.0221]
1280	0.9350	0.9438	0.9741	0.7543	+0.0136 [0.0113, 0.0157]

Note: The final column reports paired image-level bootstrap mean differences relative to the preceding resolution, with 95% percentile confidence intervals from 1,000 replicates. The observed test-set differences were +0.0221 for 800-to-960 and +0.0123 for 960-to-1280; the 1280-to-800 bootstrap mean difference was +0.0330 [0.0305, 0.0355].

Across the three primary candidate resolutions, detection performance improved as input resolution increased. The mAP50-95 increased from 0.7199 at 800 pixels to 0.7420 at 960 pixels and further to 0.7543 at 1280 pixels. The observed gain from 800 to 960 was +0.0221, whereas the gain from 960 to 1280 was +0.0123, indicating smaller marginal improvement at the highest tested resolution. The exploratory 1024-pixel check shown in Fig. 2 was not used to define an additional primary operating point because it did not receive the complete experiment matrix. The mAP50 values at 960 and 1280 were nearly saturated, with values of 0.9739 and 0.9741, respectively, indicating that the additional benefit of 1280 was more evident for stricter localization quality than for coarse detection at IoU 0.50.

The resolution-dependent gain was concentrated in mAP50-95 rather than in AP50. Bootstrap analysis (Online Resource 1, Table S12b) showed that the mean AP50 gain from 960 to 1280 pixels was +0.000865, with a 95% confidence interval of [-0.000187, 0.001886] that included zero ($P(\Delta > 0) = 0.946$). This indicates that higher resolution primarily improved stricter localization quality across IoU thresholds from 0.50 to 0.95 rather than coarse detection at IoU = 0.50, where performance was already near saturation. Consequently, the main resolution-diagnostic conclusions are drawn from mAP50-95, while AP50 is reported as a supplementary saturation indicator.

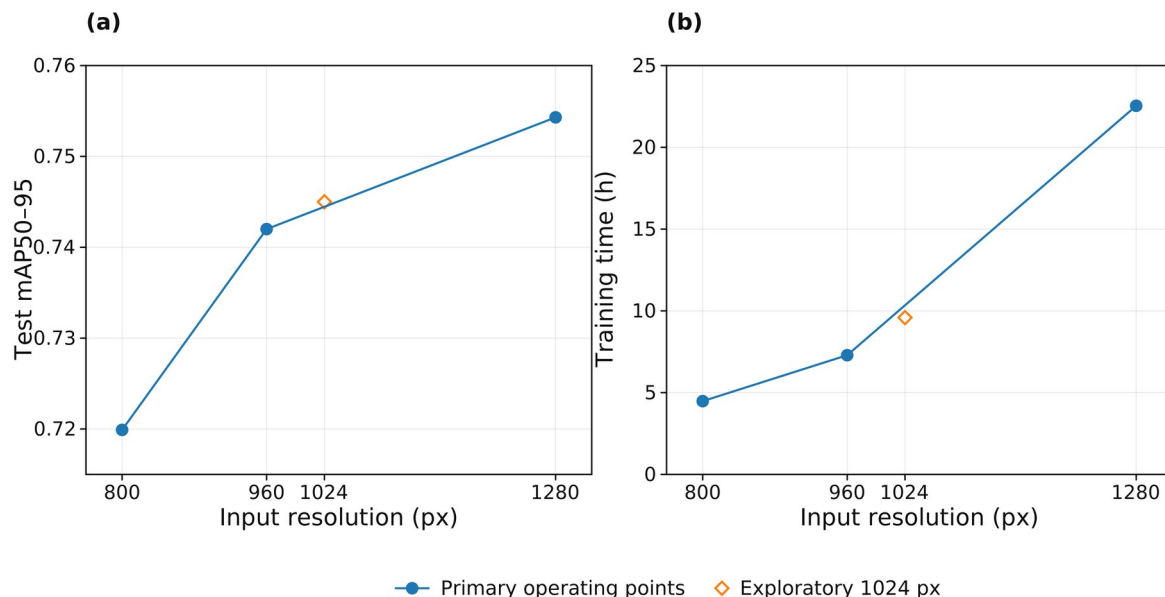


Fig. 2 Resolution-dependent strict-localization accuracy and training cost. (a) Test mAP50-95 and (b) total training time for the prespecified 800-, 960-, and 1280-pixel primary operating points. The open diamond denotes the exploratory 1024-pixel intermediate-resolution sensitivity check, which was not included in the complete primary experiment matrix.

Maximum-detection sensitivity in high-density poultry scenes

The sensitivity of mAP50-95 to the maximum number of retained detections is reported in Online Resource 1 (Table S4). Because the test images may contain hundreds of instances, a low maximum-detection cap can underestimate dense-scene performance.

The setting `max_det = 300` substantially underestimated performance in high-density images. Increasing `max_det` from 300 to 1000 improved mAP50-95 by +0.0466, +0.0495, and +0.0518 for 800, 960, and 1280 pixels, respectively. Further increasing `max_det` from 1000 to 1500 produced negligible changes. Therefore, `max_det = 1000` was sufficient for the main evaluation while avoiding an unnecessarily high detection cap.

Scale-stratified resolution effects

The scale-stratified results are visualized in Fig. 3, with exact subgroup values provided in Online Resource 1 (Table S4a). Ordinary scale-stratified AP treats the target scale group as positives and ignores ground-truth boxes from other scale groups. This analysis is a detection-accuracy diagnostic and does not define an overall operating-point ranking.

Detection performance increased with object scale. The tiny group showed substantially lower AP than the large group, confirming that small objects remain the most challenging cases in dense scenes. Increasing the input resolution from 800 to 960 improved all scale groups. Further increasing the resolution from 960 to 1280 produced smaller but still positive gains, with the large group showing the most notable improvement in this stage. These subgroup patterns concern strict-localization accuracy only; counting calibration and deployment cost are evaluated separately.

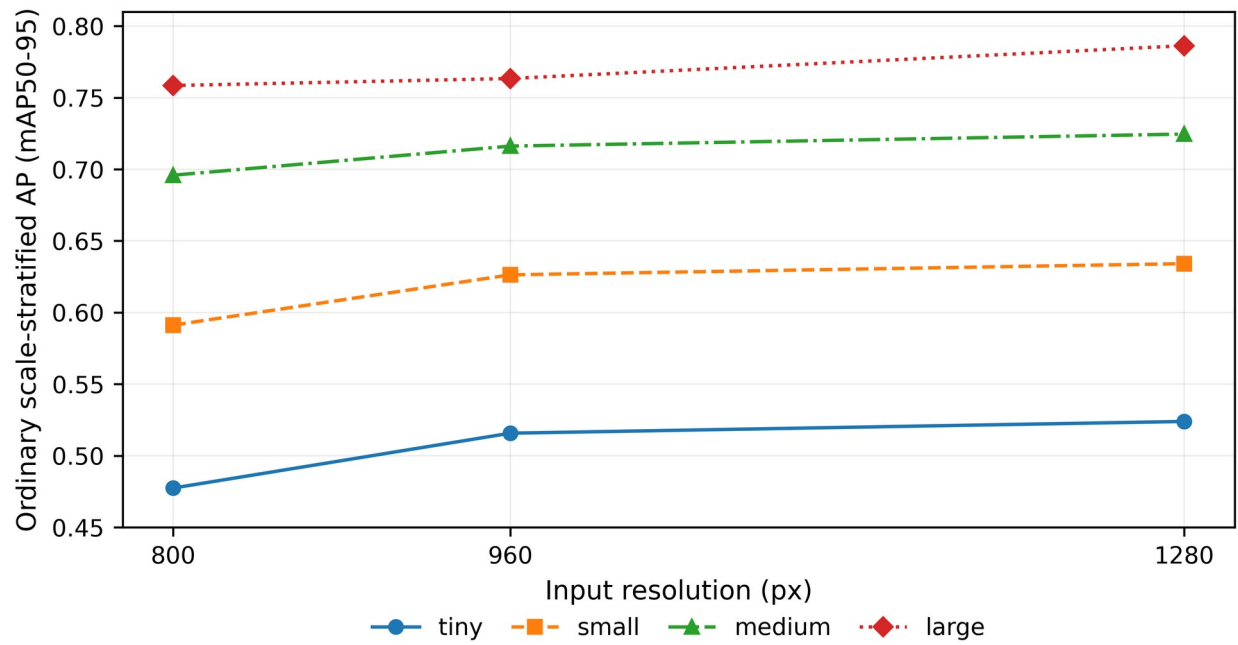


Fig. 3 Ordinary scale-stratified AP (mAP50-95) across the three prespecified primary input resolutions. Ground-truth boxes from the target scale group were treated as positives, and ground-truth boxes from other scale groups were ignored. The figure is a subgroup detection diagnostic and should not be interpreted as a universal operating-point ranking.

Density-stratified resolution effects

The density-stratified results are illustrated in Fig. 4, with exact subgroup values provided in Online Resource 1 (Table S4b). Density groups were defined by the number of ground-truth instances in the full test image.

Under ordinary density-subset AP, 1280 achieved the highest AP in all four density groups. In the high-density group, mAP50-95 increased from 0.7341 at 800 pixels to 0.7516 at 960 pixels and further to 0.7679 at 1280 pixels. In the ultra-high-density group, mAP50-95 increased from 0.7289 to 0.7570 and 0.7702. Thus, higher resolution improved this detection metric across the tested density groups, but this result does not establish a universally preferable operating point across counting and deployment objectives.

To complement this view, SGF-AP provides a contribution-oriented density perspective. In the high-density group, SGF-AP increased from 0.2299 at 800 pixels to 0.2622 at 960 pixels, whereas the 1280 result was 0.2593. In the ultra-high-density group, the corresponding values were 0.2940, 0.3170, and 0.3170. These values should not be compared numerically with the ordinary density-subset AP values in Fig. 4 or Online Resource 1 (Table S4b) because SGF-AP preserves the global ranked prediction list and does not re-optimize the ranking within each density group. Thus, the SGF-AP result does not indicate that 1280 performs worse in high-density scenes; standard global mAP and ordinary density AP still show that 1280 is the highest-accuracy setting. Instead, the SGF-AP result indicates that the additional gain from 960 to 1280 is not primarily driven by extra global-ranking support for high- or ultra-high-density groups.

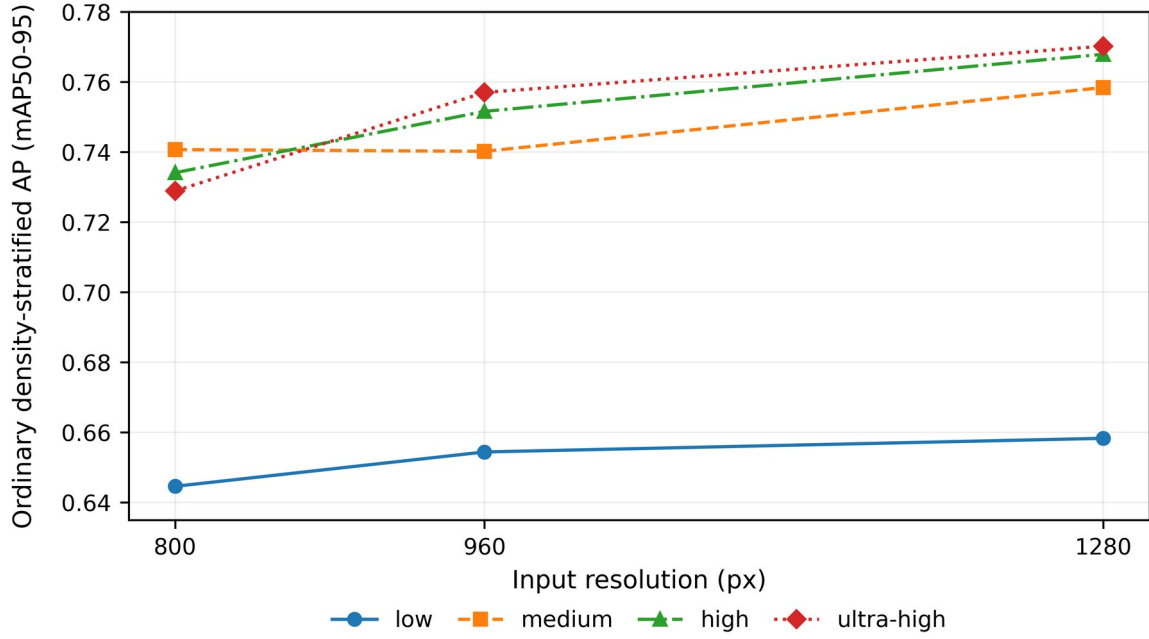


Fig. 4 Ordinary density-stratified AP (mAP50-95) across the three prespecified primary input resolutions. Density groups were defined by the number of ground-truth instances in the full test image. The plotted AP values are shown on a continuous y-axis. The figure diagnoses subgroup detection behavior only and does not define an overall resolution ranking.

TP/FN/FP diagnostics and representative dense-scene visualization

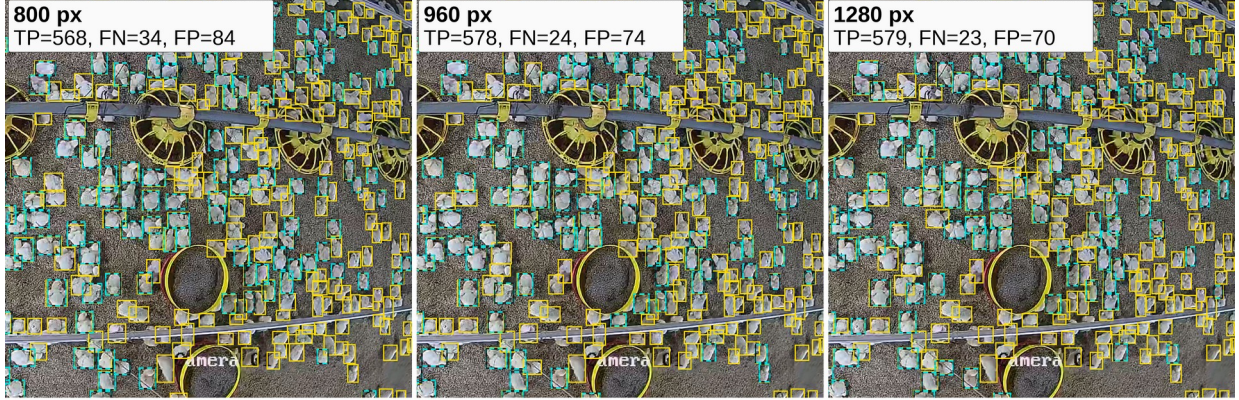
The TP/FN/FP diagnostics further clarify the density-stratified behavior. In the high- and ultra-high-density groups, 960 reduced false negatives relative to 800 and improved recall. In the high-density group, FN decreased from 621 at 800 pixels to 568 at 960 pixels. In the ultra-high-density group, FN decreased from 893 to 680. In contrast, the improvement from 960 to 1280 was more limited in TP/FN terms, while false positives were further reduced. In the ultra-high-density group, the TP/FN values for 960 and 1280 were 26,612/680 and 26,605/687, respectively, but FP decreased from 3,579 to 3,079. This suggests that the main additional benefit of 1280 over 960 in dense scenes is more related to precision and false-positive reduction than to a large recall increase.

Representative diagnostic examples are shown in Fig. 5. The visualization compares predictions under different input resolutions in high-density or ultra-high-density scenes and illustrates that resolution changes can affect false negatives and false positives differently. Ground-truth annotations and predictions are identified by the figure legend. For readability, only predictions with confidence ≥ 0.50 are displayed, while TP, FN, and FP values were computed using confidence threshold 0.25 and IoU threshold 0.50.

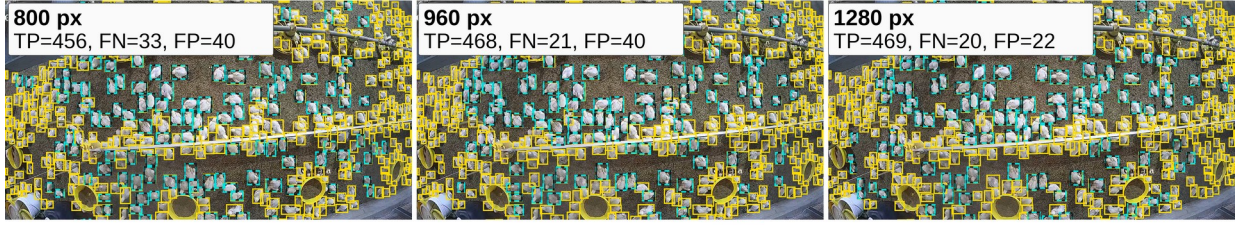
Diagnostic case A: recall-precision trade-off (571 GT instances)



Diagnostic case B: incremental improvement across resolutions (602 GT instances)



Diagnostic case C: comparable recall with fewer false positives at 1280 px (489 GT instances)



Ground truth Prediction

Fig. 5 Diagnostic qualitative comparison of YOLO11n predictions under different input resolutions in representative dense-scene crops. The rows illustrate a recall-precision trade-off, incremental improvement across resolutions, and comparable recall with fewer false positives at 1280 pixels. Ground-truth annotations and predictions are identified by the figure legend. For monochrome reproduction, ground-truth boxes use solid outlines, whereas predictions use dashed outlines. TP, FN, and FP were computed at confidence threshold = 0.25 and IoU threshold = 0.50, while only predictions with confidence ≥ 0.50 are displayed for readability. The selected crops are interpretive examples and were not used to define a generally preferred operating point. Full-image density labels, crop metadata, and visualization source data are provided in the reproducibility package. Filename prefixes such as `train__` and `val__` were inherited from the original PIO folders and do not indicate the split assignment used in this study.

Bootstrap uncertainty of resolution gains

The paired bootstrap confidence intervals are included in Table 2. The analysis used image-level paired resampling with 1,000 bootstrap replicates.

The confidence intervals did not cross zero, indicating stable mAP50-95 improvements for both resolution increases. The estimated gain from 1280 over 960 was smaller than the gain from 960 over 800, supporting the presence of diminishing marginal returns. The bootstrap delta-difference analysis is provided in Online Resource 1 (Tables S12a–S12b).

Counting calibration from detection outputs

The test-set counting results obtained using validation-calibrated thresholds are reported in Table 3. Detection AP and counting error are not equivalent because AP is a ranking-based metric, whereas counting depends on a fixed confidence threshold.

Table 3 Test counting performance using validation-calibrated confidence thresholds

Resolution	Val-selected conf	GT total	Pred total	MAE	RMSE	MAPE (%)	Bias	Rel. err. (%)
800	0.45	64,612	64,735	12.30	18.31	7.33	+0.40	+0.19
960	0.50	64,612	62,333	15.72	22.26	8.79	-7.42	-3.53
1280	0.50	64,612	63,215	14.01	20.49	7.98	-4.55	-2.16

The confidence thresholds selected on the validation set were 0.45, 0.50, and 0.50 for the primary 800-, 960-, and 1280-pixel settings, respectively. Under this adopted validation-based threshold-transfer protocol, the 800-pixel model obtained the lowest test MAE (12.30), followed by 1280 (14.01) and 960 (15.72). At the selected thresholds, the 960- and 1280-pixel models produced conservative total counts, whereas the 800-pixel threshold produced near-zero total-count bias. These results describe threshold transfer under the stated calibration procedure and do not establish that lower input resolution is generally superior for counting.

The diagnostic test-set confidence sweep in Online Resource 1 (Table S10) shows that test-tuned thresholds reduce the MAE of 960 and 1280. Fixed-threshold counting at confidence threshold 0.25 provides a complementary view: the test MAE decreased monotonically from 26.07 at 800 pixels to 24.56 at 960 pixels and 21.23 at 1280 pixels. Thus, the relative counting order changes with the threshold-selection rule. The validation threshold sweep and the exploratory intermediate-resolution sensitivity check are reported in Online Resource 1 (Fig. S3 and Tables S18-S19). Counting is therefore treated as a calibration-dependent objective rather than evidence for a universally preferable input resolution.

Latency, memory, and training-cost trade-offs

The deployment-oriented benchmark and an illustrative task-objective decision analysis are shown in Table 4. The locked-weight benchmark used the validation split, batch size 2, workers = 0, confidence threshold = 0.001, NMS IoU = 0.70, max_det = 1000, GPU FP16 inference, five warm-up repeats, and thirty formal repeats. The illustrative composite score combined main test mAP50-95 with training-time efficiency, GFLOPs efficiency, latency efficiency, and memory efficiency. It was not intended to define a universally optimal resolution. All normalization terms are defined explicitly below for the candidate resolution set $R = \{800, 960, 1280\}$.

$$N_{acc}(r) = \frac{A(r) - \min(A)}{\max(A) - \min(A)} \quad (1)$$

$$N_{cost,c}(r) = \frac{\max(C_c) - C_c(r)}{\max(C_c) - \min(C_c)} \quad (2)$$

$$Score(r) = 0.50 N_{acc}(r) + 0.20 N_{train}(r) + 0.10 N_{GFLOPs}(r) + 0.10 N_{latency}(r) + 0.10 N_{memory}(r) \quad (3)$$

In Eqs. (1)-(3), $A(r)$ denotes test mAP50-95 at resolution r , and $C_c(r)$ denotes the corresponding cost for training time, GFLOPs, latency, or memory. Accuracy is normalized as a benefit metric, whereas each cost is converted to an efficiency score so that lower cost yields a larger normalized value. The default weights define one illustrative deployment scenario in which detection accuracy is primary, training cost remains relevant under the 8 GB GPU constraint, and inference latency, GFLOPs, and memory are secondary constraints. Readers can reweight the reported components for other operational priorities. The log-cost sensitivity analysis

in Online Resource 1 (Table S20) is reported as an additional illustration of score dependence on the chosen utility model.

Table 4 Detection accuracy, computational cost, and an illustrative composite decision score

Res. (px)	Test mAP50-95	Train (h)	GFLOPs	Total latency (ms/img)	Peak reserved (MiB)	Illustrative score	Illustrative rank
800	0.7199	4.48	10.06	6.32	86	0.500	2
960	0.7420	7.29	14.49	8.52	102	0.656	1
1280	0.7543	22.54	25.76	9.15	142	0.500	2

Note: Latency and memory values were obtained from the 30-repeat locked-weight validation benchmark in Online Resource 1 (Table S14). The 800- and 1280-pixel settings had equal scores under the specified illustrative weighting and were assigned the same tie rank. The score was not intended to define a universally optimal resolution; the individual accuracy and cost values support alternative deployment priorities.

The 30-repeat locked-weight recheck showed a monotonic increase in median total latency from 6.32 ms/image at 800 pixels to 8.52 ms/image at 960 pixels and 9.15 ms/image at 1280 pixels. The 1280-pixel setting retained the highest accuracy and also required the highest GFLOPs and peak reserved CUDA memory. Under the specified illustrative composite weighting, 960 pixels obtained the highest score (0.656), whereas 800 and 1280 pixels each obtained 0.500 for different reasons: 800 minimized cost, while 1280 maximized accuracy. Moving from 800 to 960 increased training time by 62.7% and mAP50-95 by 2.21 percentage points; moving from 960 to 1280 increased training time by a further 209.2% and mAP50-95 by 1.23 percentage points. These values characterize distinct operating points rather than establishing a universal ranking.

$$Score_{\alpha}(r) = \alpha N_{acc}(r) + (1 - \alpha) [0.4 N_{train}(r) + 0.2 N_{GFLOPs}(r) + 0.2 N_{latency}(r) + 0.2 N_{memory}(r)] \quad (4)$$

The weight-sensitivity analysis in Online Resource 1 (Table S16) used Eq. (4), where α controls the emphasis on accuracy and the internal cost-efficiency mixture preserves the ratio Training:GFLOPs:Latency:Memory = 0.4:0.2:0.2:0.2. When α was below 0.340, the cost-prioritized objective selected 800 pixels; when $0.340 \leq \alpha \leq 0.652$, the stated balanced objective selected 960 pixels; and when α exceeded 0.652, the accuracy-prioritized objective selected 1280 pixels. This sensitivity analysis illustrates how the selected operating point changes with the stated objective rather than defining a generally best resolution.

Seed repeatability across three input resolutions

YOLO11n seed-repeatability results across all three input resolutions are reported in Online Resource 1 (Table S11). The main ablation used seed 42, and additional repetitions were conducted using seeds 123 and 2024. The same leakage-controlled split and evaluation protocol were maintained across all runs.

The trend remained stable across repeated runs. Test mAP50-95 ranged from 0.7199 to 0.7244 at 800 pixels, from 0.7360 to 0.7420 at 960 pixels, and from 0.7530 to 0.7559 at 1280 pixels. Therefore, the relative ordering of the three resolutions was preserved across seeds, with 1280 pixels remaining the accuracy-oriented setting and 960 pixels consistently improving over 800 pixels. The 1280-pixel seed-2024 run triggered early stopping at epoch 179 of 300, with the best model selected from epoch 129; this run was retained because its test result remained consistent with the other 1280-pixel repetitions. Because this early stop reflected a seed-specific validation-fitness plateau under patience = 50, it was not used to redefine the default training-cost value in Table

4, which consistently reports the main seed-42 training time for all resolutions. The 960-pixel seed-123 training time is a timestamp estimate because the original console log was not saved.

Auxiliary YOLO-family cross-version robustness check

To examine whether the observed resolution trend was specific to YOLO11n, YOLOv8n was additionally evaluated under the same leakage-controlled split and input-resolution protocol. This experiment served only as an auxiliary YOLO-family cross-version robustness check and was not designed to establish generality across detector families or to rank detectors.

The trend was also observed in a second lightweight YOLO variant. YOLOv8n test mAP50-95 increased from 0.7272 at 800 pixels to 0.7443 at 960 pixels and 0.7573 at 1280 pixels. This result indicates that the direction was not restricted to a single YOLO11n training run, but it does not establish generality beyond the lightweight YOLO family. The complete YOLOv8n cross-version results are reported in Online Resource 1 (Table S13).

Independent livestock-domain external validation on PigDetect

The leakage-cleaned PigDetect split provided independent livestock-domain validation of the resolution-dependent localization pattern. Table 5 combines untouched-test accuracy, paired uncertainty, and repeated deployment cost. Test mAP50-95 increased monotonically from 0.7673 at 800 pixels to 0.7757 at 960 pixels and 0.7844 at 1280 pixels. The adjacent gains were +0.0083 and +0.0087, while mAP50 remained near saturation at 0.9821, 0.9820, and 0.9830.

Table 5 Independent livestock-domain validation of resolution-dependent YOLO11n accuracy and deployment cost on the leakage-cleaned PigDetect dataset

Input (px)	Precision	Recall	mAP50	mAP50-95	Δ vs. 800	Δ vs. previous [95% CI]	GFLOPs	Latency median [IQR] (ms/img)	Peak CUDA (MiB)
800	0.9733	0.9336	0.9821	0.7673	-	-	10.06	4.76 [0.09]	68.2
960	0.9688	0.9408	0.9820	0.7757	+0.0083	+0.0083 [0.0046, 0.0125]	14.49	5.49 [0.07]	79.7
1280	0.9758	0.9407	0.9830	0.7844	+0.0171	+0.0087 [0.0046, 0.0139]	25.76	7.45 [0.16]	111.9

Note: Detection metrics were computed once on the untouched PigDetect test set (250 images and 5,436 annotated instances) using confidence = 0.001, NMS IoU = 0.70, max_det = 1000, batch size 2, and the validation-selected best.pt checkpoint. Delta intervals are 95% percentile intervals from 1,000 paired image-level bootstrap replicates. Deployment measurements were obtained on the 241-image validation split using locked checkpoints, FP16 inference, batch size 2, five discarded warm-up repeats, and thirty formal repeats. Peak CUDA memory denotes validation/inference peak allocated memory.

Paired image-level bootstrap analysis supported the mAP50-95 ordering. The 800-to-960 difference was +0.0083 with a 95% CI of [0.0046, 0.0125], and the 960-to-1280 difference was +0.0087 with a 95% CI of [0.0046, 0.0139]. Both intervals remained above zero. The corresponding mAP50 intervals crossed zero, which is consistent with the near-saturated mAP50 values. The adjacent mAP50-95 gains were similar in magnitude, indicating continued strict-localization benefit from 960 to 1280 pixels on this external dataset.

Scale-stratified analysis showed that the strongest total gain occurred for tiny instances: tiny mAP50-95 increased from 0.6277 at 800 pixels to 0.6468 at 960 pixels and 0.6602 at 1280 pixels, giving a total 800-to-1280 gain of 0.0325. The small group improved to 0.7207 at 960 pixels and then remained essentially unchanged at 0.7204, whereas medium and large groups continued to improve more modestly. Density-stratified results confirmed that denser groups were generally more difficult, but the resolution benefit was not monotonic with density. The largest 800-to-1280 gain occurred in Q3 medium-high density (+0.0263), while the Q4 high-density group gained +0.0154.

The deployment benchmark showed the expected accuracy-cost trade-off. Median total latency increased from 4.76 ms/image at 800 pixels to 5.49 ms/image at 960 pixels and 7.45 ms/image at 1280 pixels; GFLOPs increased from 10.06 to 14.49 and 25.76, and peak allocated CUDA memory increased from 68.2 to 79.7 and 111.9 MiB. Accordingly, 1280 pixels was the PigDetect accuracy-oriented setting, 800 pixels was the lowest-cost setting, and 960 pixels provided an intermediate operating point with moderate computational demand.

The locked qualitative comparison in Fig. 6 complements these aggregate results by showing a tiny-object diagnostic case, a crowded-scene diagnostic case, and a case in which the 960- and 1280-pixel predictions were broadly comparable. The figure is interpretive evidence only and does not replace the full test-set metrics or define a general operating-point recommendation.

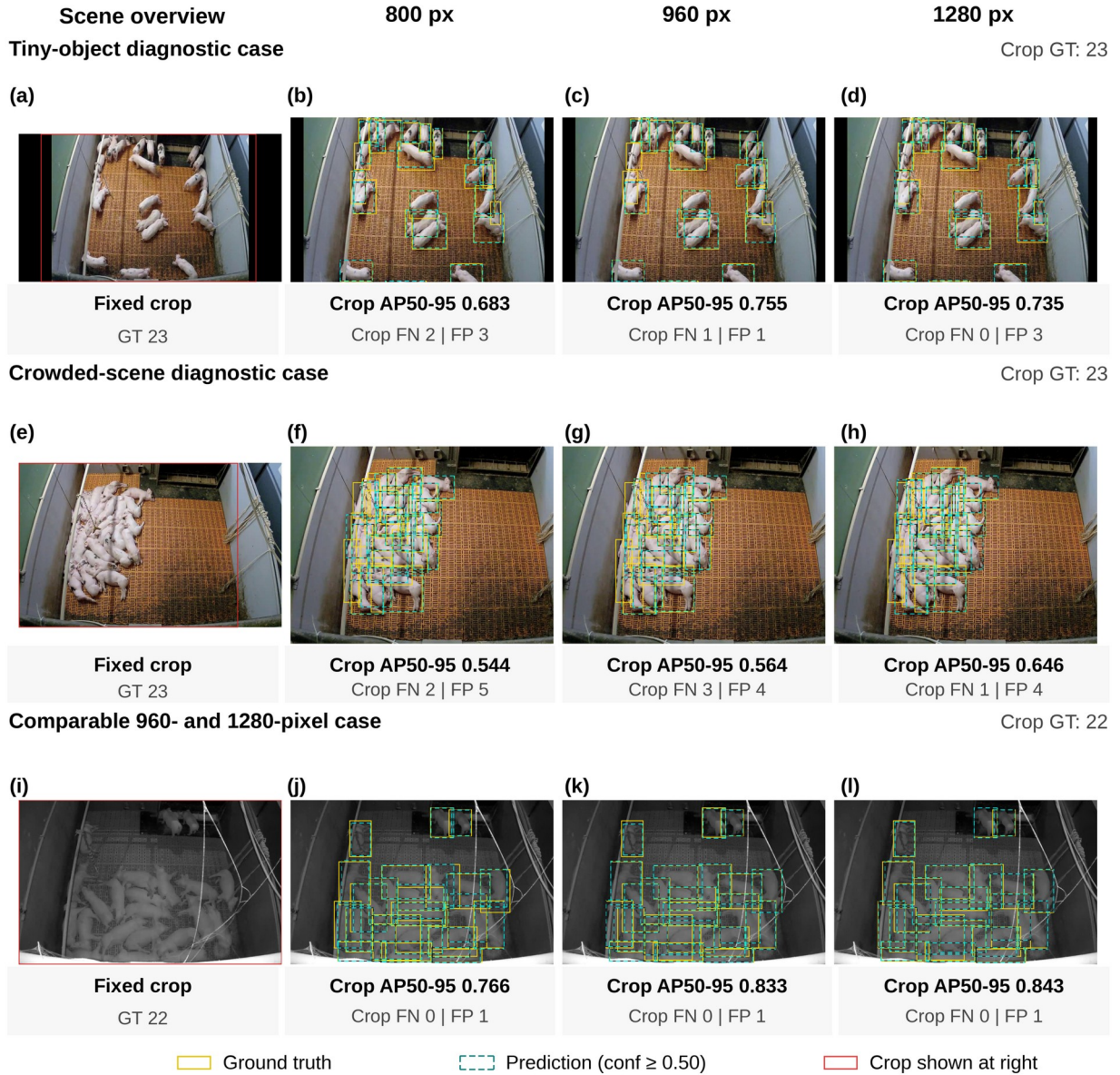


Fig. 6 Qualitative diagnostic evidence from the leakage-cleaned PigDetect test set under the locked visualization protocol. Rows show a tiny-object diagnostic case, a crowded-scene diagnostic case, and a case in which the 960- and 1280-pixel predictions are broadly comparable. Columns show the scene overview and fixed-crop predictions at 800, 960, and 1280 pixels. Ground-truth boxes, displayed predictions (confidence ≥ 0.50), and the fixed crop region are identified by the figure legend. For monochrome reproduction, ground-truth boxes use solid outlines, whereas predictions use dashed outlines. FN and FP diagnostics were computed at confidence ≥ 0.25 and IoU ≥ 0.50 . The Crop AP50-95 values beneath the crops are local diagnostic scores for the displayed regions; formal test metrics were computed on the complete test set using confidence = 0.001, NMS IoU = 0.70, and max_det = 1000. Neither the local scores nor these selected examples were used to define a universal resolution ranking.

Discussion

The principal contribution of this study is a leakage-controlled, deployment-oriented diagnosis of input-resolution trade-offs rather than identification of a universally optimal setting. Across PIO and leakage-cleaned PigDetect, increasing input resolution consistently improved strict localization, while the smaller gain from the intermediate to the highest primary resolution indicated diminishing marginal benefit.

The accuracy improvement was accompanied by higher training time, GFLOPs, latency, and memory demand. The three prespecified resolutions therefore represent objective-specific operating points: 800 pixels emphasizes lower computational demand, 960 pixels provides an intermediate accuracy-cost regime, and 1280 pixels emphasizes strict localization at the highest measured cost. The reported component values are more informative than a universal ranking because engineering priorities differ across monitoring tasks and hardware platforms.

Counting did not follow the AP ordering consistently because it was sensitive to confidence-threshold calibration. The validation-based transfer results, fixed-threshold analysis, and diagnostic sweeps should therefore be interpreted jointly as evidence that counting performance depends on calibration protocol rather than on input resolution alone.

The SGF-AP analysis provides a contribution-oriented subgroup diagnostic and is not numerically interchangeable with ordinary subgroup AP or standard global AP. Multi-seed PIO experiments supported the resolution ordering across repeated training runs, and the YOLOv8n experiment showed that the trend was also present in a second lightweight YOLO variant; this remains a YOLO-family cross-version check rather than evidence of broad detector-family generality.

Several limitations remain. The PIO validation subset was concentrated in two source components, PigDetect used one training seed, and the evidence covers two livestock datasets, one lightweight detector family, and one hardware/software stack. The three primary resolutions do not define a continuous search space, image count is only a proxy for occlusion, and alternative calibration methods or deployment platforms may shift the preferred operating point. Additional farms, camera viewpoints, detector families, and hardware environments are needed to test the transferability of the diagnostic framework.

Conclusions

This study established a leakage-controlled and deployment-oriented framework for diagnosing input-resolution trade-offs in dense broiler detection. First, higher input resolution improved strict localization on PIO and PigDetect but increased computation, memory demand, and training cost. Second, counting performance depended strongly on confidence-threshold calibration, so counting conclusions must be interpreted under the stated validation-based transfer protocol. Third, 800, 960, and 1280 pixels represented different operating objectives rather than a single ranking.

No single resolution was uniformly preferable across all objectives. The appropriate objective-specific operating point should therefore be selected from the reported accuracy, counting, uncertainty, and cost evidence according to the monitoring task and deployment constraints.

Supplementary information

The supplementary material is provided as Online Resource 1. The single PDF contains all supporting tables and figures, with an internal content map linking each evidence group to the corresponding supplementary items.

Online Resource 1 Supplementary tables and figures supporting the leakage-controlled and deployment-oriented diagnosis of input-resolution trade-offs. The file contains cross-split leakage and near-duplicate audits; source-component split diagnostics; leakage-control configuration; maximum-detection sensitivity; ordinary and SGF-AP scale/density analyses; TP/FN/FP diagnostics; counting calibration and threshold sensitivity; seed repeatability and YOLO-family cross-version robustness; paired bootstrap uncertainty and marginal-gain analysis; locked-weight deployment benchmarking; illustrative composite-score and weight-sensitivity analyses; independent PigDetect external validation; reproducibility checklist; and qualitative evidence (Tables S1-S21, including S3a, S4a-S4b, and S17a-S17g; Figs. S1-S3; and protocol information linked to main Fig. 6).

Statements and Declarations

Funding

No funding was received for conducting this study.

Competing interests

The authors declare no competing interests that are relevant to the content of this article.

Ethics approval

Not applicable to the present study. This study used existing publicly available image datasets and involved no new animal experiments, animal handling, or primary data collection by the authors.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Author contributions

Y.S. contributed to the study conception and design, data curation, split construction, leakage auditing, model training, evaluation, visualization, formal analysis, and preparation of the original draft. J.X. supervised the study and contributed to conceptual guidance and manuscript review and editing. L.H. contributed to manuscript review, discussion, and refinement. H.L. contributed to supervision, conceptual guidance, manuscript review and editing, and correspondence. Z.D. contributed to supervision, conceptual guidance, and manuscript review. All authors read and approved the final manuscript.

Data availability

The original PIO dataset is publicly available from its official source (Boniche et al. 2026; dataset DOI: 10.5281/zenodo.16686320). PigDetect is publicly available from the original dataset source associated with Henrich et al. (2026; dataset DOI: 10.25625/16UYE9). The manuscript-linked repository and archived release v1.2.1 (Song 2026) provide the PIO leakage-controlled split manifest, source-group metadata, leakage-audit summaries, group definitions, bootstrap outputs, counting-calibration outputs, evaluation scripts, traceability logs, and fixed figure source data, together with the PigDetect clean-split audit, locked test results, scale and density source tables, paired-bootstrap summary, deployment benchmark, Fig. 6 artwork and metadata, and source-to-manuscript traceability record.

These materials support inspection of the PIO split and leakage checks, input-resolution diagnosis protocol, scale- and density-stratified analyses, bootstrap analysis, counting calibration, deployment benchmarking, and the PigDetect external-validation evidence reported in Table 5, Fig. 6, and Tables S17a–S17g. Raw PIO and PigDetect images are not redistributed by the authors and should be obtained from their original dataset sources. Qualitative figure reproduction requires local access to the original images, annotations, locked checkpoints, and prediction-output files.

Code availability

PIO evaluation and figure-generation scripts, SGF-AP source data, and source records supporting the analyses in Tables S1–S20 are provided in the repository. The reproducibility checklist is reported in Online Resource 1 (Table S21). PigDetect release materials include the locked protocol record, audit summary, main-table source data, Tables S17a–S17g source files, paired-bootstrap summary, deployment benchmark, Fig. 6 artwork and

metadata, and an automated repository-alignment check. The repository does not redistribute third-party raw datasets, pretrained weights, trained checkpoints, or prediction caches.

Reproducibility checklist

A detailed reproducibility checklist for the manuscript-linked materials and public release is provided in Online Resource 1 (Table S21).

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